

JEE MAIN REVISION MASTER: PRACTICAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 111 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Practical Organic Chemistry - Core Application	111 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Practical Organic Chemistry - Core Application (Very High Yield — 111 PYQs)

Core Principle: Key problem pattern in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in al..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.481..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Evening Shift (Q#1)

A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____.
(Given Molar mass in g mol⁻¹ S: 32, BaSO₄ : 233)

- (A) 63.15%
- (B) 21.05% **[Correct]**
- (C) 48.24%
- (D) 42.10%

Correct Answer: B

Solution Analysis: Mass of compound taken = 0.314g Mass of BaSO₄ obtained = 0.4813g In BaSO₄, sulphur present per mole = 32g Molar mass of BaSO₄ = 233g mol⁻¹ So, mass of sulphur in 0.4813g of BaSO₄ is:
 $m(S) = 0.4813 \times (32 / 233)$
 $m(S) = 0.4813 \times (32 / 233) = 0.06608\text{g (approx.)}$ Percentage of sulphur in the compound:
 $\%S = (0.06608 / 0.314) \times 100 \approx 21.05\%$ Correct option: B) 21.05%